

Verifying Trigonometric Identities

Common Strategies

Verifying identities involves manipulating an equation and can often be confused with solving equations. However, the goal when verifying an identity is to work independently on either side of the equation until we can show that both sides are the same. Moving terms from one side to the other using addition, subtraction, multiplication, and division and squaring or raising both sides to any power must be avoided. There are no specific rules for verifying identities, but there are some useful strategies that can help you get started. Ultimately, this trial-and-error process requires reasoning and critical thinking, not a passive application of rules, and no shortcuts. It's typical to make several failed attempts before successfully verifying an identity.

Here are some common strategies that will help you get started.

IMPORTANT INFO: Common Strategies for Verifying Identities

There is no one method for verifying trigonometric identities; however, the following strategies may help you get started.

1. Rewrite all functions in terms of sine and cosine alone.
2. Combine ratios using a common denominator.
3. Multiply by the conjugate, then apply a Pythagorean identity.

Strategy 1 – Rewrite All Functions in Terms of Sine and Cosine Alone

Using the reciprocal and ratio identities, we can rewrite all functions in terms of sine and cosine alone. This generally allows for opportunities to simplify the expressions on either side of the equation.

EXAMPLE 1

Verify each equation is an identity.

$$\cot x \sec x = \csc x$$
$$\frac{\cancel{\cos x}}{\sin x} \left(\frac{1}{\cancel{\cos x}} \right) = \frac{1}{\sin x} \quad \checkmark$$

$\checkmark \quad \frac{1}{\sin x}$

do NOT move terms across

$$\frac{\sin \theta}{\tan \theta} \cdot \frac{\sec \theta}{\cos \theta} = \frac{\tan \theta}{\sin \theta}$$
$$\frac{\sin \theta}{\cancel{\sin \theta / \cos \theta}} \cdot \frac{1 / \cancel{\cos \theta}}{\cos \theta} = \frac{\sin \theta / \cos \theta}{\sin \theta} \quad \checkmark$$

$\checkmark \quad \frac{\sin \theta / \cos \theta}{\sin \theta}$

EXAMPLE 2

Verify the equation is an identity.

$$\cos^2 x = 1 - \tan^2 x \cos^2 x$$

$$\checkmark \quad 1 - \sin^2 x \quad \left| \quad 1 - \frac{\sin^2 x}{\cancel{\cos^2 x}} \cancel{\cos^2 x} \right.$$

$$\quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad 1 - \sin^2 x \quad \checkmark$$

Strategy 2 – Combine Ratios Using a Common Denominator

If given two ratios, combining them using a common denominator may be helpful. Similarly, at times it may be beneficial to split a single ratio into two.

EXAMPLE 3

Verify the equation is an identity.

$$\sin x (\cot x + \tan x) = \sec x$$

$$\sin x \left(\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x} \right) = \frac{1}{\cos x} \quad \checkmark$$

$$\left(\frac{\cos x}{\cos x} \right) \cos x + \frac{\sin^2 x}{\cos x}$$

$$\frac{\cos^2 x}{\cos x} + \frac{\sin^2 x}{\cos x}$$

$$\frac{\cos^2 x + \sin^2 x}{\cos x}$$

$$\checkmark \quad \frac{1}{\cos x}$$

EXAMPLE 4

Verify the equation is an identity.

$$\frac{\sec \theta}{\sin \theta} - \frac{2 \sin \theta}{\cos \theta} = \cot \theta - \tan \theta$$

$$\frac{1/\cos \theta}{\sin \theta} - \frac{2 \sin \theta}{\cos \theta} \quad \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$$

$$\left(\frac{\cos \theta}{\cos \theta}\right) \frac{1/\cos \theta}{\sin \theta} - \left(\frac{\sin \theta}{\sin \theta}\right) \frac{2 \sin \theta}{\cos \theta} \quad \left(\frac{\cos \theta}{\cos \theta}\right) \frac{\cos \theta}{\sin \theta} - \left(\frac{\sin \theta}{\sin \theta}\right) \frac{\sin \theta}{\cos \theta}$$

$$\checkmark \frac{1 - 2 \sin^2 \theta}{\sin \theta \cos \theta} \quad \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}$$

$$\frac{(1 - \sin^2 \theta) - \sin^2 \theta}{\sin \theta \cos \theta}$$

$$\frac{1 - 2 \sin^2 \theta}{\sin \theta \cos \theta} \checkmark$$

Strategy 3 – Multiply by the Conjugate then Apply a Pythagorean Identity

If the equation involves a sum or difference, multiplication by the conjugate may be helpful.

This step is usually followed by applying a Pythagorean identity. Be careful here, as the required algebra steps can sometimes be complicated.

EXAMPLE 5

Verify the equation is an identity.

Conjugates $\rightarrow (1 - \sin \theta)(1 + \sin \theta) = \frac{\cot^2 \theta}{\csc^2 \theta}$

Conjugation cancels the middle terms $\left\{ \begin{array}{l} 1 + \cancel{\sin \theta} - \cancel{\sin \theta} + \sin^2 \theta \\ 1 + \sin^2 \theta \\ \checkmark \cos^2 \theta \end{array} \right.$

$$\frac{\cos^2 \theta / \cancel{\sin^2 \theta}}{1 / \cancel{\sin^2 \theta}} \left(\frac{\sin^2 \theta}{\sin^2 \theta} \right)$$

$$\cos^2 \theta \checkmark$$

EXAMPLE 6

Verify the equation is an identity.

$$\frac{\sin^2 x}{1 - \cos x} = 1 + \cos x \left(\frac{1 - \cos x}{1 - \cos x} \right)$$

$$\checkmark \quad \frac{1 - \cos^2 x}{1 - \cos x} \quad \frac{1 - \cancel{\cos x} + \cancel{\cos x} - \cos^2 x}{1 - \cos x}$$

$$\frac{1 - \cos^2 x}{1 - \cos x} \quad \checkmark$$

EXAMPLE 7

Verify the equation is an identity.

$$\frac{1}{\sec x + \tan x} = \frac{1 - \sin x}{\cos x} \left(\frac{1 + \sin x}{1 + \sin x} \right)$$

$$\left(\frac{\cos x}{\cos x} \right) \frac{1}{\frac{1}{\cos x} + \frac{\sin x}{\cos x}} \quad \frac{1 - \sin^2 x}{\cos x (1 + \sin x)}$$

$$\checkmark \quad \frac{\cos x}{1 + \sin x} \quad \frac{\cancel{\cos^2 x}}{\cancel{\cos x} (1 + \sin x)}$$

$$\frac{\cos x}{1 + \sin x} \quad \checkmark$$

